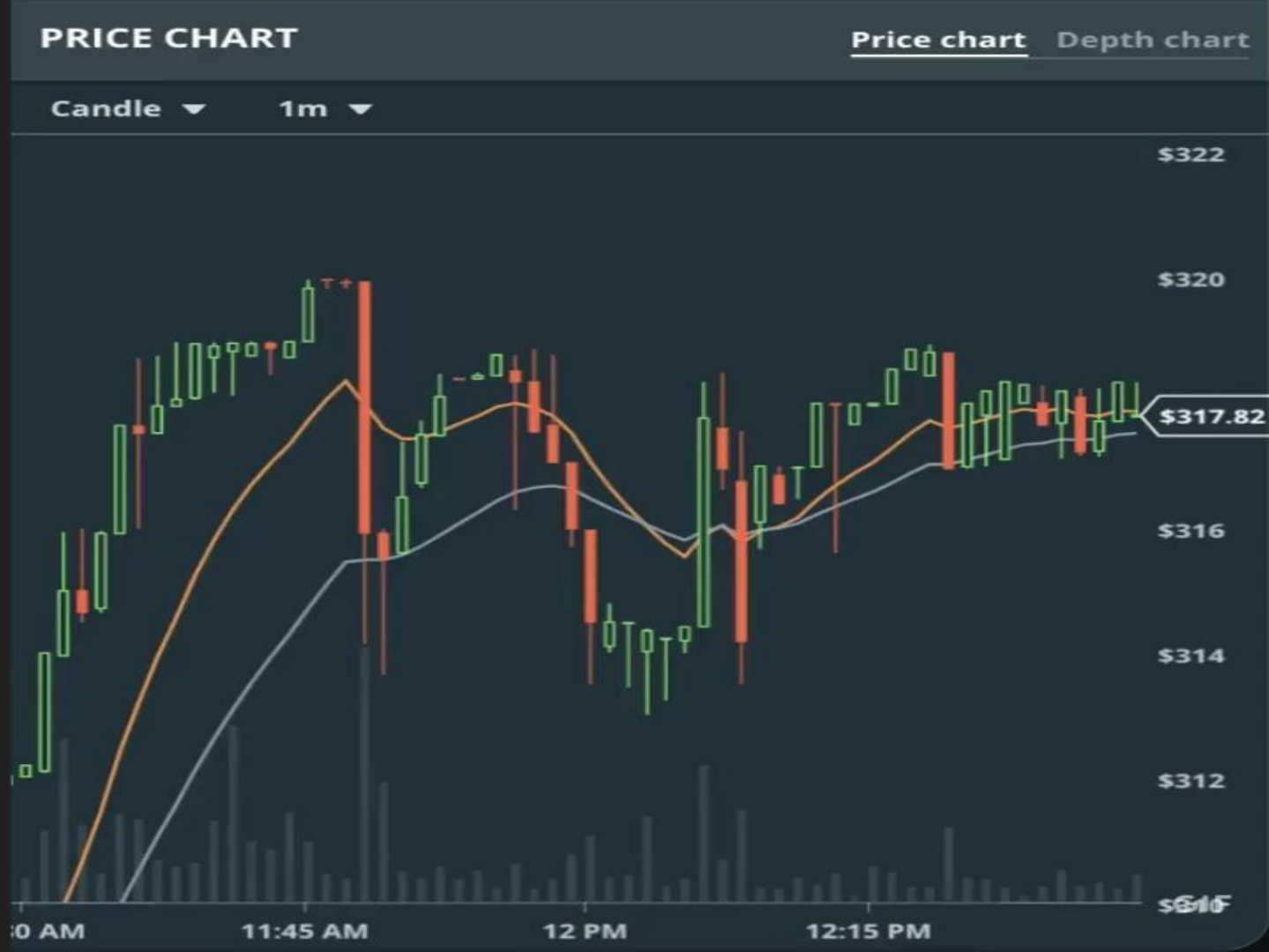


Mitigating Flash Crashes in DeFi

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The Problem



Traditional Market Protections

Traditional equity markets have developed two regulatory mechanisms to prevent flash crashes:

- **Market-Wide Circuit Breakers (MWCBS):** Implemented in 1988 as a response to Black Monday (Dow dropped 22%)
- **Limit Up-Limit Down (LULD):** Began in 2013 due to the 2010 Flash Crash (hundreds of stocks fell to \$0.01 before recovering to normal within minutes)

Market-Wide Circuit Breakers (MWCBS)

- **Trigger:** Three levels based on S&P 500 single day decline (7%, 13%, 20%)
- **Action:** 15-minute halt (Levels 1 & 2) or close for day (Level 3)
- **Scope:** Halts trading across all U.S. equity and derivative exchanges simultaneously
- **Usage:** Triggered 5 times ever. 4 of them in March 2020 during COVID volatility
 - Bitcoin from 2020-2025 would have triggered **84** Level 1, **20** Level 2 and **4** Level 3 MWCBS

Limit Up-Limit Down (LULD)

- **Trigger:** Individual stock limit orders reaching $\pm 5\%$ or $\pm 10\%$ bands from reference price
- **Action:** 5-minute halt if price doesn't stabilize
- **Scope:** Applies to stocks individually
- **Usage:** Triggered tens to hundreds of times per day

Traditional Approaches Fail in DeFi

- DEXs don't answer to a central authority/regulator
- AMMs don't use resting limit orders
- ...

A Crypto-Native Solution

Rather than attempting to control exchanges, embed protection **directly into token contracts**

- **Token-level control:** Circuit breaker lives in the token itself
- **Works across DEXs and chains:** Protection follows the token wherever it's traded on-chain
- **Decentralized:** No central operator needed

Token Contract Implementation

```
uint public haltUntil;

function activateCircuitBreaker(haltDuration) {
    haltUntil = block.timestamp + haltDuration;
    emit CircuitBreakerActivated(haltUntil);
}

modifier whenNotCircuitBreakerActive() {
    if (block.timestamp < haltUntil) {
        revert CircuitBreakerActive();
    }
    _;
}

// modifier applied to transfer and transferFrom
```

When to Activate?

Problem: How do we detect a flash crash and trigger the circuit breaker?

Three initial ideas, each with downsides:

- **External:** Off-chain service monitors prices—efficient but not trustless or crypto-native
- **On Every Transfer:** Token contract checks price before each transfer—slows execution and increases gas costs
- **Periodic Checks:** Token contract checks price at intervals—creates temporal blind spots

Preferred Approach: Monitoring Contract

Solution: Separate on-chain monitoring contract that alerts the token contract

Architecture:

- Monitoring contract checks price feed(s)
- When threshold exceeded, calls token contract to activate circuit breaker
- Token contract only handles halt logic

Why this is better:

- On-chain and transparent
- No additional gas costs for users
- Constant monitoring

Cross-Chain Protection

Problem: Tokens exist on multiple chains

Crypto-native solution: Use cross-chain messaging protocols

- Circuit breaker on chain A detects crash
- Broadcasts halt signal to chain B, chain C, ...
- All chains halt transfers as soon as they receive the halt signal
- Can also have independent triggers on each chain

Open question: Which chains should have monitoring contracts?

Who Controls It?

Problem: Who decides thresholds, halt duration, and other parameters?

Crypto-native solution: DAO governance

- Token community votes on all circuit breaker parameters
- Parameters can be updated as market conditions evolve
- DAO treasury funds monitoring contract
 - Incentives aligned: token holders have the most to lose, so they pay for protection

Additional Benefits

- **No venue changes needed:** AMMs and DEXs don't need to modify anything
- **New venues automatically covered:** When the token is listed on a new DEX, protection works immediately
- **Bidirectional protection:** Can halt on extreme upside moves too
- **Customizable per token:** Each DAO sets thresholds based on their token's risk profile (stablecoin vs. memecoin)
- **Complex trigger logic:** Beyond simple % drops, can consider market-wide volatility, time-of-day patterns, etc. (higher gas costs for complexity)

Limitations

- **No utility during halt:** Token can't be used for payments, governance, staking, or other purposes during the halt
- **Ongoing gas costs:** DAO treasury must continuously fund monitoring contract operations
- **Price feed speed:** If price feeds are too slow then protection may activate too late to be effective
- **Doesn't work on CEXs:** Out of scope for this work

Next Steps

- Determine price feed source
- Research trigger thresholds through historical data analysis
- Implement and validate the idea through simulations
- Quantify impact metrics such as crash frequency, price integrity, and user protection